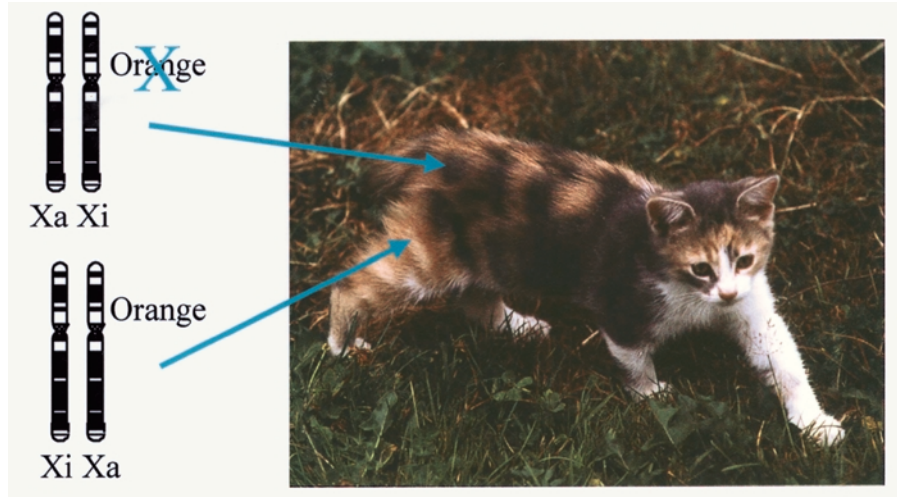




The tortoiseshell or calico cat is an example of mosaicism. This is caused by random X chromosome inactivation. A female has two X chromosomes. If she is heterozygous, one will carry the black allele, and one the orange allele. Random inactivation (X_i) yields a population of cells with orange fur, and another population of cells with black fur. (The white colored fur seen in calico cats (that distinguishes them from tortoiseshell cats) is not associated with the X chromosome inactivation, rather, it is due to an autosomal gene.)

are mosaic. Health sometimes depends on what fraction of the cells in a tissue express the functional gene.

For reasons that are not yet well understood, some females exhibit non-random inactivation patterns. If the chromosome with the normal copy of a gene is inactivated in most of the cells in a female's body, and if the normal protein is vital for some function, the female is likely to develop a disease.

With some X-linked diseases, cells that contain a mutation on the active X chromosome proliferate less during development than cells that carry the mutation on the inactive X chromosome. In such cases, the female primarily expresses the normal gene.

Unlike females, males with an X-linked mutation will usually show signs of the disease, because they have no second functional copy. (Females will usually show symptoms if they inherit a mutated copy of the gene from each parent.) Males therefore inherit X-linked diseases, such as Duchenne muscular dystrophy, hemophilia, or colorblindness, much more commonly than females. Some X-linked disorders are almost never found in males, which may seem paradoxical until we consider that the absence of a functional gene can be so harmful that most males who inherit the disease die before being born.

Such is the case with Rett syndrome, an X-linked, dominant neurological disorder. This disorder is due to a mutation in a gene called *MECP2*. The disorder is primarily found in females, whose mosaicism gives them partial protection from its effects. Only a handful of males with Rett syndrome are known.

Other Types of Mosaicism

Since humans consist of more than ten trillion cells, it is not surprising that mutations occur in the genes in some of these cells, rendering the individ-

ual a mosaic. In some cases such changes have limited impact and are found in only a few cells. In other cases they may lead to cancer or disease.

We are all likely to have some cells in our body that have acquired mutations, and therefore everyone may be considered a mosaic at some level. Mosaicism can create differences among a person's cells. It can also result in differences between "identical" twins.

Most females are mosaics, due to X chromosome inactivation, though their mosaicism does not necessarily involve any disease gene. Chromosomal and mitochondrial mosaicism are also observed frequently.

Chromosomal Mosaicism. Every time a cell divides, the genetic material assembled in chromosomes needs to be divided too. If the chromosomes do not separate evenly, then cells are formed with missing or extra chromosomes. Such cells are called aneuploid. If this occurs early in embryonic development, a significant proportion of cells in an individual will be abnormal.

Chromosomal mosaicism may also result from the "rescue" of a fertilization that resulted from an aneuploid sperm or egg. If a fertilized egg contains three copies of a particular chromosome, a condition called trisomy, instead of the normal two, one of the extra copies can be "lost" if the chromosomes divide unevenly, restoring the normal chromosome number to the daughter cell.

Typically, in fetuses surviving the first trimester of pregnancy, the abnormal cells are found in placental but not in fetal tissues. Cells with three copies of a chromosome may be able to survive better in placental tissues, or there may be stronger selection against the growth of such cells in fetal tissues.

Trisomy is occasionally associated with pregnancy complications, such as poor fetal growth, but it may be common in placental tissues, and mosaicism confined to the placenta has been suggested to occur in up to 5 percent of births. Trisomic cells can also be found in the fetus itself, although this occurs much more rarely. Sometimes the abnormal cells will be present in only one type of tissue, such as the skin or lungs. Such variability has made it difficult to determine how often such chromosome mosaicism occurs and how it affects the health of an individual.

Mitochondrial Mosaicism. The mitochondria are **organelles** in the cytoplasm that release energy stored in molecules for cells to use. They contain their own small chromosome. The mitochondrial chromosome contains 16,569 base pairs, compared with the nuclear chromosomes, which, together, contain three billion base pairs.

Two features make mitochondria prone to mosaicism. First, their DNA is mutated more frequently than the nuclear DNA, in part because of the more dangerous cellular environment facing mitochondria and in part because mitochondria are not equipped to repair mutations as effectively as the nucleus.

Second, each mitochondrion contains numerous copies of its genome, and there are thousands of mitochondria in each cell. Thus individuals can have mutations in some of their mitochondrial genomes that are not found in their other mitochondria. This can lead to variable expression of diseases

trisomy presence of three, instead of two, copies of a particular chromosome

organelles membrane-bound cell compartments

associated with mitochondrial mutations. Deletions of part of the mitochondrial genome appear to accumulate in different tissues with age and have been suggested to be a critical factor in normal human aging. SEE ALSO CHROMOSOME, EUKARYOTIC; DNA REPAIR; GENE EXPRESSION: OVERVIEW OF CONTROL; MITOCHONDRIAL DISEASES; MITOCHONDRIAL GENOME; MUSCULAR DYSTROPHY; NONDISJUNCTION; X CHROMOSOME.

Carolyn J. Brown

Bibliography

Avner, Philip, and Edith Heard. "X-Chromosome Inactivation: Counting, Choice and Initiation." *Nature Reviews: Genetics* 2 (2001): 59–67.

Lyon, Mary F. "Gene Action in the X-Chromosome of the Mouse (*Mus musculus* L.)." *Nature* 190 (1961): 372–373.

Internet Resource

Online Mendelian Inheritance in Man. Johns Hopkins University, and National Center for Biotechnology Information. <<http://www.ncbi.nlm.nih.gov/omim>>.

Mouse See *Rodent Models*

Muller, Hermann

Geneticist
1890–1967



Hermann Muller.

Hermann Joseph Muller was one of the founding members of the “fly lab” that was initiated by Thomas Hunt Morgan. In the early part of the twentieth century, this lab was the center of important research into the role of chromosomes in inheritance, using the fruit fly *Drosophila* as a model organism in experiments. The major members included Morgan, Alfred Henry Sturtevant, Calvin Blackman Bridges, and Muller, all working at Columbia University around 1910 to 1915, when their major contributions to classical genetics were carried out. Muller was the only one of Morgan’s students to also win a Nobel Prize.

Muller’s career was unusual in that he worked in several countries. He was a third-generation American, but he left the United States in 1932 to work in Germany, the Soviet Union, and Edinburgh, Scotland, before returning to his homeland. He had a productive career as an experimental and theoretical geneticist, but he also had a life-long interest in seeing genetics applied to society. This made him controversial and sometimes put his life in jeopardy.

The first phase of Muller’s career was spent at Columbia, where the fly lab was established, and at Rice University, where he held his first tenured position from 1915 to 1922. During this phase he did work that contributed to the understanding of crossing over and gene mapping (discovered by Morgan and Sturtevant). Muller also clarified the meaning of genetic mutation by limiting the concept to variations in the individual gene. He proposed the gene as the basis of life, arguing that only genes had the property of replicating their errors, essential for the evolution of life. He was the first to measure mutation rates, and he designed stocks of *Drosophila* to detect them.

Muller's second phase was the decade he spent at the University of Texas, from 1922 to 1932. During this time he was heavily committed to the study of mutation, culminating in his Nobel Prize work on the induction of mutations by **radiation**. Muller rapidly followed up his initial reports and founded a new field of radiation genetics. He also used X rays as a tool to delete chromosomes, and used these small deleted chromosomes to reveal the mechanism of genetic functions such as **dosage compensation**, a phenomenon he was first to interpret. Muller showed that mutations are produced in proportion to radiation dosage and that chromosome rearrangements (such as **translocations**) were induced at higher dosages.

In his third phase, which ran from 1932 to 1940 and which he spent working in Berlin, the Soviet Union, and Scotland, Muller studied chromosome structure, gene structure, and changes in gene function when genes were moved from their normal chromosome location. He also introduced the idea that genes arise from preexisting genes, when he discovered that a fly mutation called Bar eyes arose from a physical duplication of genes. In his last phase, Muller was back in the United States (from 1940 to 1967), working mainly at Indiana University, where he worked out the mechanism of cell death from radiation exposure and calculated the amount of mutation normally occurring in humans (genetic load) each generation. He became a critic of the Cold War policies of the United States, favoring a strong nuclear defense as a protection against Stalinism but also calling for mutual treaties to limit nuclear arms. He also fought hard against the misuse of radiation by health practitioners and industry.

In addition to his fundamental work on fly genetics, Muller contributed to human genetics through studies of twins that he conducted in the 1920s. He argued that the relation of observable character traits to genes is very complex, a problem he had first studied in detail in *Drosophila*, when he investigated the verifiability of shape and size in the "truncate" and "beaded" wing mutations. Muller stressed that an observable trait such as intelligence or longevity will be influenced by many genes as well as by the environment, and that simple one-gene/one-trait relationships were the exception rather than the rule in complex organisms.

Muller felt that advocates of **eugenics** programs ignored environmental modifiers and the complex residual heredity that he called modifier genes. He denounced the American eugenics movement in 1932 at the Third (and last) International Congress of Eugenics. Yet he remained an idealist about eugenics, favoring a positive eugenics based on "germinal choice," a non-coercive way for educated people to choose the genetic character of their own children. SEE ALSO EUGENICS; FRUIT FLY: *DROSOPHILA*; EVOLUTION OF GENES; MAPPING; MORGAN, THOMAS HUNT; MUTATION; TWINS.

Elof Carlson

Bibliography

Carlson, Elof. *The Gene: A Critical History*. Philadelphia: Saunders Publishing Co., 1966.

Muller, Hermann J. "The Development of the Gene Theory." In *Genetics in the Twentieth Century*, L. C. Dunn, ed. New York: Macmillan, 1951.

Olby, Robert. *The Path to the Double Helix*. Seattle: University of Washington Press, 1974.

radiation high energy particles or waves capable of damaging DNA, including X rays and gamma rays

dosage compensation equalizing of expression level of X-chromosome genes between males and females, by silencing one X chromosome in females or amplifying expression in males

translocations movements of chromosome segments from one chromosome to another

eugenics movement to "improve" the gene pool by selective breeding

loci sites on a chromosome (singular, locus)

homologous carrying similar genes

antigen a foreign substance that provokes an immune response

polymorphism DNA sequence variant

nucleotides the building blocks of RNA or DNA

Multiple Alleles

Alleles are alternative forms of a gene, and they are responsible for differences in phenotypic expression of a given trait (e.g., brown eyes versus green eyes). A gene for which at least two alleles exist is said to be polymorphic. Instances in which a particular gene may exist in three or more allelic forms are known as multiple allele conditions. It is important to note that while multiple alleles occur and are maintained within a population, any individual possesses only two such alleles (at equivalent **loci** on **homologous** chromosomes).

Examples of Multiple Alleles

Two human examples of multiple-allele genes are the gene of the ABO blood group system, and the human-leukocyte-associated **antigen** (HLA) genes.

The ABO system in humans is controlled by three alleles, usually referred to as I^A , I^B , and I^O (the “I” stands for isohaemagglutinin). I^A and I^B are codominant and produce type A and type B antigens, respectively, which migrate to the surface of red blood cells, while I^O is the recessive allele and produces no antigen. The blood groups arising from the different possible genotypes are summarized in the following table.

Genotype	Blood Group
$I^A I^A$	A
$I^A I^O$	A
$I^B I^B$	B
$I^B I^O$	B
$I^A I^B$	AB
$I^O I^O$	O

HLA genes code for protein antigens that are expressed in most human cell types and play an important role in immune responses. These antigens are also the main class of molecule responsible for organ rejections following transplantations—thus their alternative name: major histocompatibility complex (MHC) genes.

The most striking feature of HLA genes is their high degree of **polymorphism**—there may be as many as one hundred different alleles at a single locus. If one also considers that an individual possesses five or more HLA loci, it becomes clear why donor-recipient matches for organ transplantations are so rare (the fewer HLA antigens the donor and recipient have in common, the greater the chance of rejection).

Polymorphism in Noncoding DNA

It must be realized that although the above two are valid examples, most genes are not multiply allelic but exist only in one or two forms within a population. Most of the DNA sequence variation between individuals arises not because of differences in the genes, but because of differences in the noncoding DNA found between genes.

An example of a noncoding DNA sequence that is extremely abundant in humans is the so-called microsatellite DNA. Microsatellite sequences consist of a small number of **nucleotides** repeated up to twenty or thirty times.

For instance, the microsatellite composed of the dinucleotide AC is very common, appearing about one hundred thousand times throughout the human genome.

The interesting feature about microsatellites is that they are very highly polymorphic for the number of repeat lengths. For example, one particular individual might possess the microsatellite sequence ACACACACAC at a specific locus on one chromosome, and the sequence ACACACACACACACACAC at the same locus on the other homologous chromosome.

Making Use of Polymorphic DNA

Multiple alleles and noncoding polymorphic DNA are of considerable importance in gene mapping—identifying the relative positions of genetic loci on chromosomes. Gene maps are constructed by using the frequency of crossing-over to estimate the distance between a pair of loci. To obtain a good estimate, one must analyze a large number of offspring from a single cross. In laboratory organisms such as the fruit fly *Drosophila*, programmed crosses can be carried out so it is possible to use gene loci to construct a reliable genetic map. In humans, this is not the case. For this reason, the more highly variable noncoding regions are of considerable importance in human genetic mapping. SEE ALSO BLOOD TYPE; IMMUNE SYSTEM GENETICS; MAPPING; POLYMORPHISMS; TRANSPLANTATION.

Andrea Bernasconi

Bibliography

Alberts, Bruce, et al. *Molecular Biology of the Cell*, 4th ed. New York: Garland, 2002.
Strachan, Tom, and Andrew P. Read. *Human Molecular Genetics*. New York: Bios Scientific Publishers, 1996.

Muscular Dystrophy

Muscular dystrophies (MDs) are a group of disorders that share three characteristics: They are inherited, they cause progressive weakness and muscle wasting, and the primary defect is localized to skeletal muscle, sparing the nerves. Although selected limb muscles develop some degree of weakness in all dystrophies, to distinguish among the different types, it is critical to know the mode of inheritance, the age of onset, and whether muscles other than limb muscles are also affected. For example, some dystrophies additionally affect eye and lip closure; another type affects eye movement ability, as well as swallowing and speech.

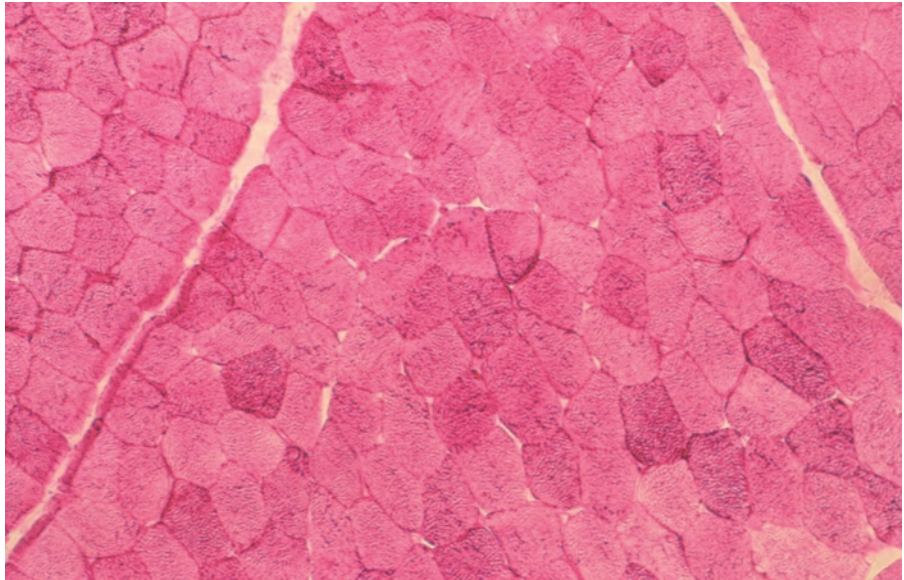
More than thirty types of MDs are now recognized. Three of the more prevalent forms—Duchenne, myotonic, and limb-girdle dystrophies—will be discussed from the standpoint of the presenting symptoms, age of onset, inheritance pattern, causative genes, and the availability of prenatal and presymptomatic molecular testing.

Duchenne Muscular Dystrophy

Duchenne muscular dystrophy (DMD) is an X-linked recessive disorder with a worldwide occurrence of one in four thousand newborn males, with approximately one-third of the cases arising from new mutations. DMD was



A cross-section of healthy muscle tissue.

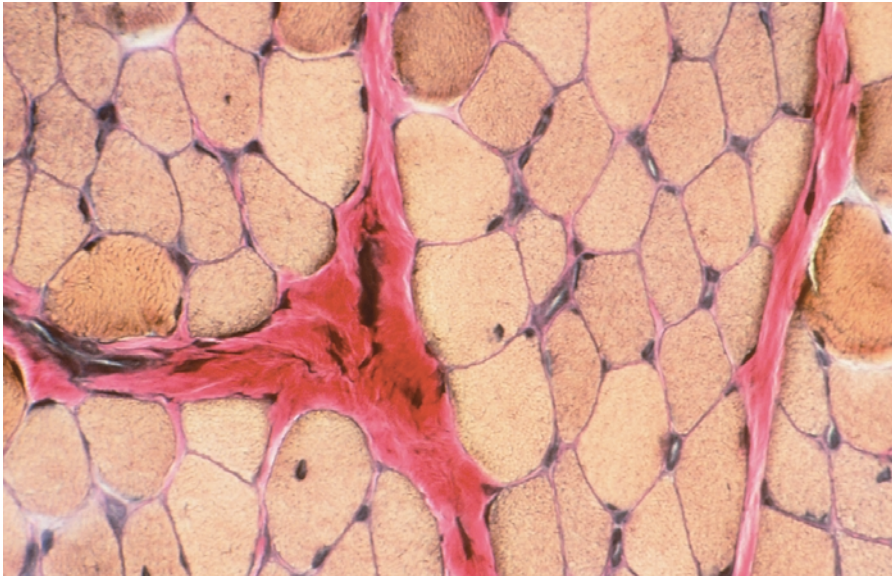


named after the French neurologist Duchenne de Boulogne, who described the disorder in 1861. Becker muscular dystrophy (BMD), named after Peter Becker, a German geneticist who first described it in the mid-1950s, is a disorder that is very similar to DMD but has a much milder course. In 1983 these disorders were first shown to be located on the short arm of the X chromosome. The disorders are now known to be allelic, meaning an alternate form of the DMD gene causes BMD.

Because DMD is X-linked, almost all cases occur in males. Boys with DMD are normal at birth, and their early motor milestones occur at normal times. The manifestations of DMD are frequently apparent from the time they begin to walk, due to the developing weakness of the hip-girdle and upper-leg muscles. Their gait is unsteady and clumsy, resulting in frequent falls. If running is attempted, it is slow and waddling. The calf muscles often are enlarged enough to be termed “hypertrophic,” implying that these children are muscular and strong. In reality calf *pseudohypertrophy* is present: When the calves are examined microscopically, the amount of muscle tissue is markedly reduced, having been replaced by fat and fibrous tissue.

As the disorder progresses, the Achilles tendons tighten, causing toe-walking, which further compromises patients’ gait and balance. Stair-climbing, rising from a fall, and even walking on level ground becomes more arduous. Even if they undergo Achilles tendon lengthening surgery or use leg braces, virtually all DMD boys require a wheelchair for mobility before the age of thirteen. Weakening of the muscles of the upper extremities and neck and of the respiratory muscles occurs in parallel to that of the lower extremities, although at a slower rate. By some time in their twenties, if not before, nearly all DMD patients will die, often due to an overwhelming respiratory infection resulting in respiratory failure, cardiac arrest, or both.

The causative gene for DMD (named *dystrophin*) and the protein product (also named dystrophin) were identified in 1986. Dystrophin is found on the inner side of the membrane that surrounds skeletal muscle fibers (the sarcolemma). It is usually absent or severely deficient in DMD boys, and this causes the sarcolemma to weaken and develop tears, allowing excess cal-



A cross-section of Duchenne muscular dystrophy tissue. Note the increase in connective tissue and the resulting decrease in healthy muscle tissue.

cium to enter the muscle fiber. This eventually leads to the death of muscle fibers, and, when a sufficient number of fibers are involved, muscle weakness results.

The *dystrophin* gene is the largest known human gene, encompassing two thousand kilobases (two million bases) of genomic DNA. In 55 percent to 65 percent of DMD or BMD cases, large deletions of the *dystrophin* gene can be found. Duplications within the gene account for about 5 percent of cases. DNA testing, available through a number of commercial laboratories in the United States, is based on the identification of these large deletions and duplications. DNA tests can confirm a diagnosis of DMD or BMD, and they can be used for accurate **carrier** or prenatal testing.

Myotonic Muscular Dystrophy

Myotonic muscular dystrophy (DM, or dystrophia myotonica) is the most common adult-onset muscular dystrophy, having a frequency of one per twenty thousand persons in the general population. Myotonia, the delayed relaxation of a voluntary muscle after it is contracted, and muscle weakness are the hallmarks of the disorder. For example, a person with DM using a hammer will not immediately be able to release his grip on the handle when finished. It is an **autosomal** dominant disorder, but there is great variability in the disorder's severity and in the number of manifestations it leads to.

A unique feature of this dystrophy is a genetic phenomenon called pleiotropy, or multisystem involvement, despite the single genetic defect. The potential involvement includes multiple organs and organ systems other than skeletal muscle, including the cardiac, respiratory, gastrointestinal, central nervous, endocrine, and dermatologic systems, as well as bone or eyes. A congenital variety occurs in which infants are born floppy, often require respiratory assistance, have extremity deformities, and are both physically and mentally retarded.

Patients often initially complain of a loss of hand strength (they have difficulty twisting off caps from bottles, for example) or of tripping while

carrier a person with one copy of a gene for a recessive trait, who therefore does not express the trait

autosomal describes a chromosome other than the X and Y sex-determining chromosomes



walking or climbing stairs, due to the weakness of muscles that extend the feet and toes. Weakness may progress to involve the shoulder and hip girdles and, in some cases, is severe enough to necessitate the use of a wheelchair. Droopy eyelids, wasting of facial and neck muscles, and frontal balding frequently occur, producing atypical facial appearance.

The gene for DM is a protein kinase gene (known as *DMPK*) and is located on the long arm of chromosome 19. The disorder arises from a repeated sequence of three nucleotides—cytosine (C), thymine (T), and guanine (G)—in the gene. Individuals without DM have C-T-G repeats that contain between 5 and 37 iterations of the triplets. By contrast, repeats that are between 40 and 170 iterations long are found in the mild phenotype, repeats between 100 and 1,000 iterations are found in the “classic phenotype,” and repeats of between 500 and 3,000 are found in the congenital phenotype. A number of laboratories in the United States perform this triplet repeat assay for diagnostic, prenatal, and presymptomatic testing.

Limb-Girdle Muscular Dystrophy

Limb-girdle muscular dystrophy (LGMD) has been described both as a heterogeneous group of disorders and as a diagnosis of exclusion. Any patient who has weakness of the shoulder and hip-girdle muscles and who otherwise has been excluded from the other MDs will be diagnosed with LGMD. Using the patterns of inheritance that exist within LGMD, a classification system has been created to simplify the heterogeneity: Both autosomal dominant (LGMD1) and recessive families (LGMD2) are well recognized. The number of LGMD genes that have already been identified has further improved the classification.

The frequency of LGMD in the general population is reported to be one in twenty-five thousand. The age at onset may vary widely. In some individuals, onset is in early childhood and in others it occurs in the forties and fifties, but most commonly it occurs in the teens to early adulthood. The characteristic pattern of muscle involvement is symmetric weakness, beginning initially in the hip and shoulder girdle, but usually noticed in the hips before the shoulders. Thus slowness in running, difficulty rising from a low seat, and difficulty ascending stairs are all common complaints from affected individuals. As LGMD progresses, it will involve upper-leg and arm muscles and may eventually affect the muscles that extend the feet and wrists. Lower-extremity weakness may become severe enough to require a wheelchair.

Among the LGMD1 types, five have chromosomal linkages, but only in one is the protein product of the gene known. LGMD2 is better characterized, with nine chromosomal localizations, five with known proteins. Almost all these proteins are membrane-associated proteins (just as dystrophin is). When they are abnormal in structure or deficient in quantity, they affect the stability of the muscle membrane, resulting in the same pathological process that was described for DMD. Commercial testing in the United States is only available for some of the LGMD2 types and is performed using muscle tissue.

Treatment of the Muscular Dystrophies

As of mid-2002, gene therapy treatment of LGMD was tried in a very small number of patients. These early experiments delivered a functional gene to

a very small muscle in the foot and were designed to test the long-term safety and effectiveness of the treatment. Gene therapy for DMD is much more problematic, because of the immense size of the gene and the distribution throughout the body that would be required for effective treatment. Drug treatment with prednisone or other corticosteroids is being used, although at best this provides another six to twelve months of mobility before a wheelchair becomes necessary. There are no effective treatments for myotonic dystrophy as of 2002, although research continues in many laboratories worldwide. SEE ALSO GENE THERAPY; GENETIC TESTING; INHERITANCE PATTERNS; PRENATAL DIAGNOSIS; TRIPLET REPEAT DISEASE.

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Bibliography

Emery, Alan E. H., ed. *Neuromuscular Disorders: Clinical and Molecular Genetics*. Chichester, U.K.: John Wiley & Sons, 1998.

———. *Muscular Dystrophy, The Facts*. Oxford, U.K.: Oxford University Press, 2000.

Hoffman, Eric P. "Muscular Dystrophy: Identification and Use of Genes for Diagnostics and Therapeutics." *Archives of Pathology and Laboratory Medicine* 123 (1999): 1050–1052.

Internet Resource

Muscular Dystrophy Association. <<http://www.mdausa.org>>.

Mutagen

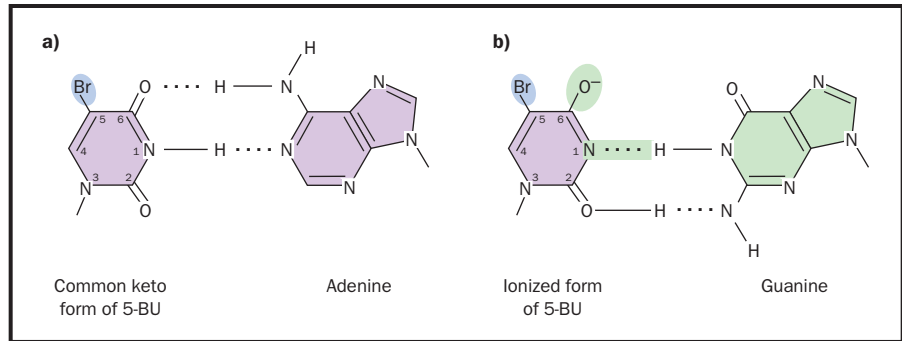
A mutagen is any substance or agent that can cause a mutation, or change in the sequence or structure of DNA. Mutagens are classified on the basis of their physical nature and the types of damage they do. A mutagen is not the same as a carcinogen. Carcinogens are agents that cause cancer. While many mutagens are carcinogens as well, many others are not. The Ames test is a widely used test to screen chemicals used in foods or medications for mutagenic potential.

Chemical Mutagens

There are many hundreds of known chemical mutagens. Some resemble the bases found in normal DNA; others alter the structures of existing bases; others insert themselves in the helix between bases; while others work indirectly, creating reactive compounds that directly damage the DNA structure.

"Base analogs" are molecules whose chemical structure is similar to one of the four DNA bases (adenine, thymine, cytosine, and guanine). Because of this similarity, they can be incorporated into the helix during DNA replication. A key feature of mutagenic base analogs is that they form base pairs with more than one other base. This can cause mutations during the next round of replication, when the replication machinery tries to pair a new base with the incorporated mutagen. For instance, 5-bromodeoxyuridine (5BU) exists in two different forms. One mimics thymine and therefore pairs with adenine during replication, while the other mimics cytosine and therefore pairs with guanine. In its thymine-mimicking form, 5BU can be incorporated across from an adenine. If it then converts to its cytosine-like form, during the next round of replication, it will cause a guanine to enter the opposite strand, rather than the correct adenine.

The mutagen 5-bromo-deoxyuridine exists in two forms, one of which pairs with adenine, and the other of which pairs with guanine. Adapted from <http://fig.cox.miami.edu/Faculty/Dana/baseanalog.jpg>.



organic composed of carbon, or derived from living organisms; also, a type of agriculture stressing soil fertility and avoidance of synthetic pesticides and fertilizers

“Base-altering mutagens” cause chemical changes in bases that are part of the DNA. For example, nitrite preservatives in food convert to the mutagen nitrous acid. Nitrous acid causes deamination, or loss of an -NH_2 group, of cytosine. When this occurs, cytosine becomes uracil, a base that is not normally incorporated in DNA but that is very similar to thymine. Unless repaired, this uracil will cause an adenine to enter the opposite strand instead of a guanine. Many base-altering mutagens are complex **organic** molecules. These are formed in large quantities in smoke, making up the “tar” of cigarette smoke, for example. They act as alkylating agents, combining with DNA to form bulky groups that interrupt replication.

“Intercalating agents” are flat molecules that insert themselves between adjacent bases in the double helix, distorting the shape at the point of insertion. Where this occurs, DNA polymerase may add an additional base opposite the intercalating agent. If this occurs in a gene, it induces a frameshift mutation (that is, it alters the reading of the gene transcript, changing which amino acids are added to the encoded protein). Ethidium bromide is one such agent, widely used in DNA research because its dark color allows DNA to be easily visualized. This is useful in gel electrophoresis, for instance, to find the DNA bands that have been separated in a gel.

Other damaging agents include chemicals that create “free radicals” inside a cell. Free radicals are compounds in which an atom, usually an oxygen, has an unbonded electron. Free radicals are highly reactive and can cause several types of damage to DNA.

Light and Radiation

Radiation refers to two different phenomena: light and high-energy particles. Visible light represents a small slice of the electromagnetic spectrum, which includes long-wavelength (low-energy) radio waves and short-wavelength (high-energy) ultraviolet waves, plus X rays and gamma rays. These high-energy forms can directly disrupt DNA by breaking its chemical bonds. In severe cases, this can break apart chromosomes, causing chromosome aberrations. More often, they create mutagenic free radicals in the cell. X rays were first used by Hermann Muller to induce mutations in fruit flies. They continue to be used to create mutations in model organisms to study genes and their effects.

Ultraviolet light is less energetic than X rays but causes mutations nonetheless. The higher-energy form, UV-B, is more toxic than UV-A, because of its potential to cause cross-linking between adjacent thymine or

cytosine bases, creating a so-called pyrimidine dimer (cytosine and thymine are chemically classified as pyrimidines). Pyrimidine dimers interrupt replication. UV-A does not cause dimer formation but can still cause mutations by creating free radicals.

Another meaning of the term “radiation” is high-energy particles released during the breakdown of radioactive elements, such as uranium. These particles are either electrons (called beta particles) or helium nuclei (called alpha particles). Their energy is sufficient to disrupt DNA structure, or to create free radicals.

Repairing the Damage

DNA is constantly being damaged, and it is constantly being repaired as well. It is only when the damage is not repaired that a mutation can lead to cancer or cell death. The DNA repair enzymes can recognize damaged nucleotides and remove and replace them. The human liver contains a large number of enzymes whose role is to detoxify toxic compounds, mutagenic or otherwise, by chemically reacting them. However, in some cases these enzymes (called cytochrome P450s) actually create mutagens during the course of these reactions. Such “bioactivation” may be a significant source of mutagens. SEE ALSO CARCINOGENS; CHROMOSOMAL ABERRATIONS; DNA REPAIR; MULLER, HERMANN; MUTAGENESIS; MUTATION.

Richard Robinson

Bibliography

Philp, Richard B. *Ecosystems and Human Health: Toxicology and Environmental Hazards*, 2nd ed. Boca Raton, FL: Lewis Publishers, 2001.

Brusick, David. *Principles of Genetic Toxicology*, 2nd ed. New York: Plenum Press, 1987.

Mutagenesis

Mutagenesis is the process of inducing mutations. Mutations may occur due to exposure to natural **mutagens** such as ultraviolet (UV) light, to industrial or environmental mutagens such as benzene or asbestos, or by deliberate mutagenesis for purposes of genetic research. For geneticists, the study of mutagenesis is important because mutants reveal the genetic mechanisms underlying heredity and gene expression. Mutations are also important for studying protein function: Often the importance of a protein cannot be characterized unless a mutant can be made in which that protein is absent.

Noninduced Mutagenic Agents

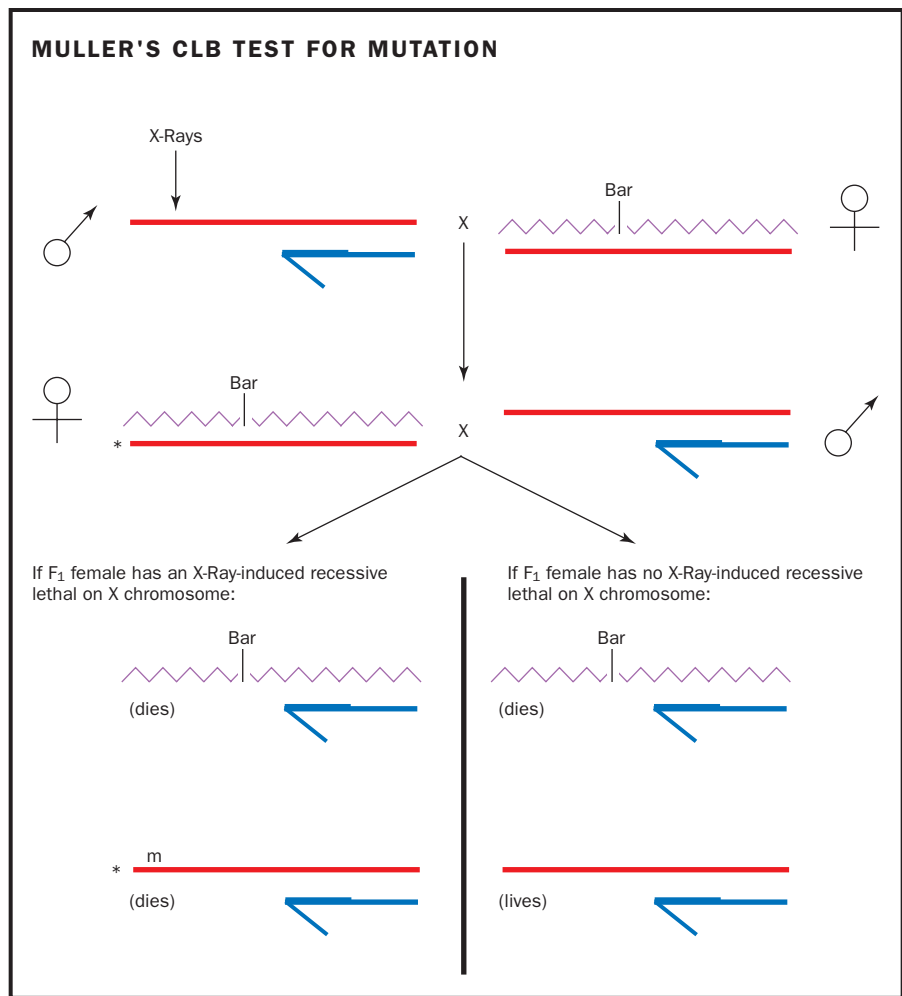
Environmental agents can influence the mutation rate not only by increasing it, but also by decreasing it. For example, antioxidants, which are found commonly in fruits and vegetables, are thought by many to protect against mutagens that are generated by normal cellular respiration. In addition to protective agents, however, many plants also contain **deleterious** mutagens known as **carcinogens**. Many chemical mutagens exist both naturally in the environment and as a result of human activity. Benzo(a)pyrene, for example, is produced by any incomplete burning, whether of tobacco in a cigarette or of wood in forest fires.

mutagen any substance or agent capable of causing a change in the structure of DNA

deleterious harmful

carcinogens substances that cause cancer

Figure 1. First the parental males were treated and mated to females carrying the *CIB* X chromosome. Second, those female offspring with Bar eyes were mated individually so that only their offspring would be present in the bottle. Third, after the offspring hatched, the bottle was examined for the presence of males. There were no males with the *CIB* X chromosome because the *l* gene kills them. So, any bottles with no males represent one lethal mutation on the other X chromosome, which originated in the treated male.



recessive requiring the presence of two alleles to control the phenotype

Spontaneous (noninduced) mutations are very rare, and finding them is difficult because most are **recessive**. The recessive nature of most mutations means that they will not be evident in most of the individuals who inherit them, for they will be hidden by the presence of the dominant allele. The rarity of mutations means that many individuals must be examined to find a mutant, whether they are people, other organisms, or even cells in culture.

Creating Mutations

To overcome the problem of the rarity of mutations, researchers induce mutations with a variety of agents. Hermann Muller was the first to do this when, in 1927, he used X rays on fruit flies (*Drosophila*) to increase the mutation rate by more than 100-fold. Other high-energy forms of radiation can also be used to create mutations.

The first chemical to be recognized as a mutagen was mustard gas, which had been developed during World War I, but not tested until World War II by Auerbach and Robson, at the University of Edinburgh. Since then a wide variety of chemicals have been discovered that are also mutagenic. Some induce mutations at any point in the cell cycle, by disrupting DNA structure. Others only act during DNA replication. Called base

analogs, these latter chemicals have structures similar to the bases found in DNA, and are incorporated instead of the normal base.

Transposable genetic elements (also called transposons, or “jumping genes”) can also induce mutations. These elements insert randomly into the genome, and may disrupt gene function if inserted into a gene or its promoter. Finding the organism with a disrupted gene is made easier if the transposon carries with it a reporter gene whose product can be identified, or a selectable marker that allows the transformed cells to live while non-transformed ones die. (The use of reporter genes and selectable markers are techniques used in genetic analysis in the laboratory.) The transposon sequence itself serves as a molecular tag. Thus, if the target gene (the gene being studied) is interrupted, finding the transposon allows the researcher to find the gene.

All of the above methods disrupt genes randomly. However, specific genes can also be targeted, for “site-directed mutagenesis,” if their sequence is known. Using the known sequence, a matching DNA sequence is inserted into a single-stranded **vector**. Short, complementary, partial sequences containing the desired mutation are then synthesized. These are allowed to pair up, and DNA polymerase is then used to complete the complementary strand. Further replication **amplifies** the number of copies of the mutant. In bacteria, the mutant gene can be placed on a plasmid for transformation of the bacteria. The bacteria make the mutant protein, and the effect of the mutation can then be studied. This is a key tool in studying how amino acid sequences affect protein structure, since individual amino acids can be changed, one at a time.

In eukaryotes, the mutant gene can be inserted into the chromosome of an experimental organism by “homologous recombination,” a system in which the mutant gene switches places with the normal chromosomal gene. Such techniques can “knock out” and “knock in” genes bearing the desired mutations.

The First Mutagenesis Assay

Before DNA sequencing became widespread, most mutations could only be detected by their effects on the **phenotype** of the organism. Many mutations are recessive, however, and do not affect the phenotype if present in only one allele. Hermann Muller, who pioneered the study of mutations, overcame this problem by focusing on the X chromosome in his studies of the genetics of *Drosophila*, the fruit fly. While females have two X chromosomes, males have only one, so any mutated gene carried on the X chromosome is expressed in males, even if it is recessive. Hence recessive lethal mutations on the X chromosome kill any male inheriting them, but would not kill a female. Muller’s method examined all of the genes on the X chromosome that could mutate to give a recessive lethal mutation. Muller used X rays to generate mutants. X rays are a very high-energy form of radiation, and break the DNA at numerous points. The method is shown in Figure 1.

Muller treated adult males with X rays and mated them to females who carried one copy of a specially prepared X chromosome, called *CIB*. This chromosome had a gene to prevent crossing over (*C*), which kept the chromosome intact; a lethal recessive gene (*I*) to kill any males that inherit it;

Michael Smith of Canada received the 1993 Nobel Prize in physiology or medicine for invention of site-directed mutagenesis. He shared the prize with Kary Mullis, who invented the polymerase chain reaction.

vector carrier

amplifies multiplies

phenotype observable characteristics of an organism

and a dominant “bar eye” gene (*B*) that resulted in a distinctive phenotypic change, making it easy to find female flies that inherited it.

Muller mated X-ray treated males with *CIB*-carrying females. All female offspring from this crossbreeding received one treated X chromosome from the male (which might or might not have carried a lethal recessive gene). They also received one X from the female, either normal or *CIB*. He selected only the bar-eyed females for further mating. To determine which of these females carried an X-ray induced lethal recessive, he separated each female into a separate jar, and examined their offspring.

Three types of males were created in this cross, depending on what type of X chromosome they inherited. Males inheriting the *CIB* chromosome died, due to the presence of the *l* gene. Males inheriting an X-ray-treated X chromosome with a lethal recessive died. Males inheriting an X-ray-treated X chromosome without a lethal recessive lived. Therefore, any jar with live males indicated that the mother did not carry a lethal recessive. Any jars with no males indicates the mother carried a lethal recessive, originally induced in the X-ray-treated male. The analysis was rapid because an experienced person could examine a bottle of flies and see at a glance if there were males present. Subsequent studies showed that this method tested almost 1,000 genes simultaneously, thus making it practical to use when detecting rare mutations. Unfortunately the breeding takes quite a lot of time, so this assay has now fallen largely into disuse, despite its historical importance.

Detecting Mutations

Today the mutagenic potential of chemicals is considered in evaluating the mutagenic risks posed by chemical exposure. Many new methods have been developed to determine if chemicals to which people will be exposed, such as new drugs, food additives, and pesticides, are mutagens. Since mutations can occur in any organism, and because there are many different kinds of mutations, there are a correspondingly wide variety of tests to detect them. No one test detects them all.

The Ames test was the first and remains the only test to be almost universally required by regulatory agencies as a minimum standard for determining if a chemical is mutagenic. The test is conducted in *Salmonella* bacteria. Since bacteria have only one chromosome, recessive mutations can be detected readily. Rare mutations are easily detected because mutants can be selected very simply. Several variants have been added to the original test, allowing for detection of many types of mutations. In an effort to make the test more relevant to human risk, one variant uses an extract of liver to mimic the biochemical modifications of chemicals that occur in the human liver.

Assays for Chromosome Aberrations. Chromosomal aberrations can be detected by examining cells in mitosis or meiosis for changes (see Figure 2). Typically, bone marrow cells of mice or rats are examined for *in vivo* tests. Any cells can be used for tests of cells in culture, but Chinese hamster cells or human **fibroblasts** are most commonly used. Another test, called the micronucleus test, is also commonly used. Micronuclei are small nuclei that arise from pieces of chromosomes or whole chromosomes that have been lost during cell division. They are conveniently detected in mouse red blood

fibroblasts undifferentiated cells that normally give rise to connective tissue cells

cells, which have no normal nucleus but which often retain micronuclei. The micronucleus assay is also widely used in cultured cells.

Assays for Somatic Mutations. Recessive mutations can be detected more readily on a mammalian X chromosome than on the other chromosomes, because only one X chromosome is active. Therefore, detection of the mutagenic potential of a substance in mammals can be most efficiently performed by analyzing the X-linked mutations. A system using the X-linked gene *hprt* has been widely used because the enzyme is not essential and because the addition of the drug thioguanine kills all cells except mutants. A count of the cells that can be cultured in the presence of thioguanine is a count of *hprt* mutants. SEE ALSO AMES TEST; MUTAGEN; MUTATION; MUTATION RATES.

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Bibliography

Griffiths, Anthony J. F., et al. *An Introduction to Genetic Analysis*. New York: W. H. Freeman, 2000.

Muller, Hermann J. "Artificial Transmutation of the Gene." *Science* 66 (1927): 84–87.

Rubin, G. M., and A. C. Spradling. "Genetic Transformation of *Drosophila* with Transposable Element Vectors." *Science* 218 (1982): 348–353.

Internet Resource

United Nations Scientific Committee on the Effects of Atomic Radiation. <<http://www.unscear.org/>>.

Mutation

A mutation is any heritable change in the genome of an organism. For a population, heritable mutations provide the source of genetic variation, without which evolution could not occur: If all individuals of a species were genetically identical, every subsequent generation would be identical regardless of which members of the species reproduced successfully. For an individual organism, mutations are rarely beneficial, and many cause genetic diseases, including cancer. For researchers, mutations (either spontaneous or introduced) provide important clues about gene location and function.

Phenotypic Effects and Evolution

Mutations in the **germ-line** cells are heritable and provide the raw material upon which natural selection operates to produce evolution. Mutations in **somatic** cells, which are cells that are not germ line, are not heritable but may lead to disease in the organism possessing them.

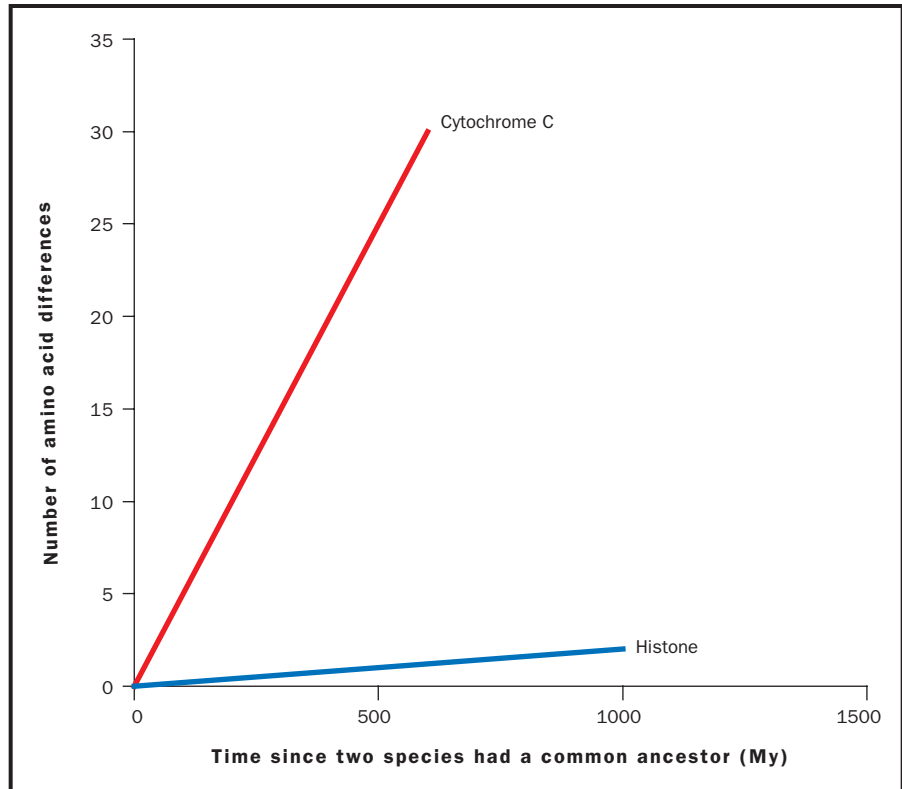
Most mutations do not cause disease and are said to be "silent" mutations. This is for at least two reasons. First, most DNA does not code for genes, so changes in the sequence do not affect the types or amounts of protein made and there is no change in the **phenotype** of the organism. Second, most sexually reproducing organisms are diploid, meaning they possess two copies of every gene. Many types of mutation simply disable one copy, leaving the other intact and functional. Therefore these mutations display a recessive inheritance pattern, with no effect on phenotype unless an individual inherits two copies of the mutation. Diploid species can accumulate a large pool of such recessive mutations, which are mostly disadvantageous and thus contribute to the burden of genetic disease.

germ-line cells giving rise to eggs or sperm

somatic nonreproductive; not an egg or sperm

phenotype observable characteristics of an organism

Figure 1. Since there is dated fossil evidence for the species used to generate the line, the evolutionary history of species without fossil evidence can be estimated from the number of amino acid changes (substitutions) in the protein.



allele a particular form of a gene

Some mutations lead to detrimental alterations of the normal phenotype and are, therefore, selected against. Very occasionally, the mutant phenotype is superior and provides a selective advantage, which leads to an increase in the frequency of this mutant **allele** and, thus, to evolution of the population. Alternatively, a disadvantageous mutation in one environment may become advantageous in another, again leading to increased frequency of this allele.

Molecular Basis of Mutations

DNA is composed of a double helix, each side of which is a long string of four types of nucleotides. Each nucleotide possesses identical sugar-phosphate groups that contribute to the DNA backbone but differs in the structure of the base suspended between the two backbones. The bases are adenine, thymine, cytosine, and guanine (A, T, C, G). Because of their structure, A pairs only with T across the double helix, and C only with G.

Within genes, the sequence of DNA encodes a sequence of amino acids used to build a protein. The DNA is read in triplets of bases, with each triplet coding for an amino acid. With the recognition that the genetic information lies in the sequence of bases in the DNA, it became possible to understand the chemical nature of gene mutations and how these could be as stable as the original allele of the gene.

Consideration of the genetic code linking DNA and amino acids reveals how mutations can either alter a protein, have no effect, or prevent it from being produced entirely. Mutations fall into four broad categories (point mutations, structural chromosomal aberrations, numerical chromosomal

Deletion of the **T** in the DNA **TAC CAT CAT CAT CAT CAT CAT ...**
 becomes **TAC CAT CAC ATC ATC ATC ATC ATC ...**

Figure 2. Frameshift mutations are deletions or insertions of bases in the DNA. Unless a multiple of three is involved in the deletion, all of the subsequent codons are altered, as illustrated by the deletion of T above. Hence, different amino acids will appear in the mutant protein, and its function will be destroyed.

aberrations, and transposon-induced mutations), each of which may be subdivided further.

Point Mutations

“Point mutations” are small changes in the sequence of DNA bases within a gene. These are what are most commonly meant by the word “mutation.” Point mutations include substitutions, insertions, and deletions of one or more bases.

If one base is replaced by another, the mutation is called a base substitution. Because the DNA is double-stranded, a change on one strand is always accompanied by a change on the other (this change may occur spontaneously during DNA replication, or it can be created by errors during DNA repair. Consequently, it is often difficult to know which base of the pair was mutated and which was simply the result of repairing the mismatch at the mutation.

For example, the most common mutation in mammalian cells is the substitution of a G-C pair with an A-T pair. This could arise if G is replaced by A and subsequently the A is replicated to give T on the other strand. Alternatively, the C could be replaced by a T and the T could then be replicated to give an A on the complementary strand, the final result being the same. It is believed that the G-C to A-T conversion most commonly begins with a C-to-T mutation. This is because most of these mutations occur at DNA sequences in which C is methylated (i.e., chemically modified by the addition of a $-\text{CH}_3$ group). The methylated form of C can be converted to a base that resembles T (and thus pairs with A) by removal of an $-\text{NH}_2$ group (deamination)—a relatively common event.

Base substitution mutations are classified as transitions or transversions. Transitions are mutations in which one pyrimidine (C or T) is substituted by the other and one purine (G or A) is substituted on the complementary strand. The G-C to A-T conversion is a transition mutation, since C becomes T.

Transversions are mutations in which a purine is replaced by a pyrimidine or vice versa. Sickle cell anemia is caused by a transversion: T is substituted for A in the gene for a hemoglobin subunit. This mutation has arisen numerous times in human evolution. It causes a single amino acid change, from glutamic acid to valine, in the β subunit of hemoglobin. Sickle cell anemia was the first genetic condition for which the change in the protein was demonstrated in 1954 by Linus Pauling (a Nobel laureate from the California Institute of Technology) and subsequently shown to be a single amino acid difference by Vernon Ingram (a Nobel laureate from the Massachusetts Institute of Technology).



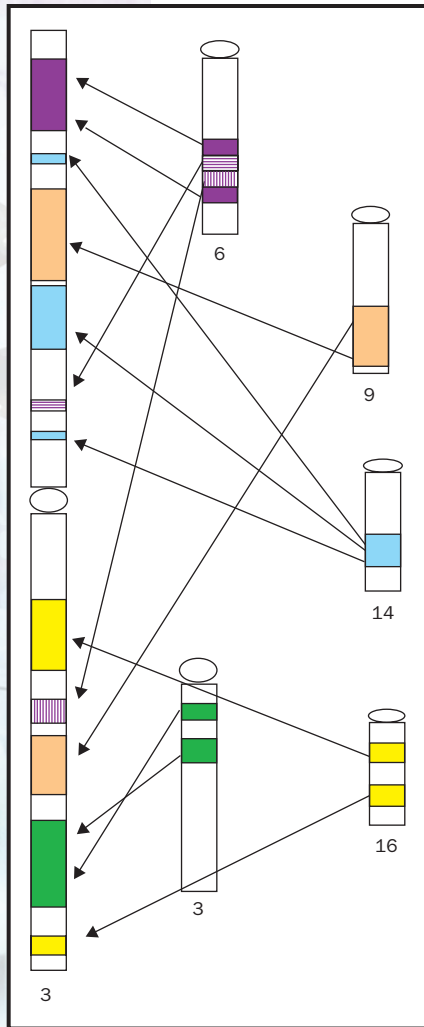


Figure 3. An example of the relationship between human and mouse chromosomes. Human chromosome 3 is on the left, and five mouse chromosomes are on the right. Although there have been numerous changes, these blocks of hundreds of genes have remained together for hundreds of millions of years in both species, and are not randomly arranged.

codon a sequence of three mRNA nucleotides coding for one amino acid

Base substitutions are sometimes silent mutations—mutations that do not change the amino acid sequence in the protein encoded by the gene. Silent mutations are possible because the original and mutated sequences can code for the same amino acid, given the redundancy of the genetic code. In the divergence between sea urchins and humans, for example, one of the histone proteins has only two amino acid substitutions, although the gene has many base pair substitutions. Histones are proteins around which DNA is wrapped in chromosomes. The very close similarity in sequence between such distantly related organisms is an indication of how critical the structure is for the function: Most mutations that change it are very disadvantageous.

One type of substitution mutation that almost always inactivates the gene is mutation to a stop **codon**. A stop codon ends the assembly of the protein, and a truncated protein is usually not active biochemically. Many recessive genetic diseases occur when a mutation converts a coding triplet to a stop codon.

Other mutations involve the insertion or deletion of one or more base pairs in the DNA. When they occur in genes, such mutations typically inactivate the encoded protein, because they change the “reading frame” of the gene. The DNA sequence is translated in groups of three nucleotides. Insertion or deletion of a nucleotide changes the sets of triplets, and thus every subsequent amino acid is altered, changing the protein completely, as shown in Figure 2. Stop codons also frequently arise from insertions or deletions.

Naturally occurring trinucleotide repeat sequences (e.g., CAGCAG CAGCAG) are hot spots for certain important human mutations that involve the insertion of more copies of the repeated sequence. For example at the locus for Huntington’s disease, a sequence of 10–29 copies of CAG is normal and stable, but if there are 30–38, there is a high rate of mutation to increased numbers of copies, and if there are 39 or more copies, middle-age dementia called Huntington’s disease results.

Functional Consequences and Inheritance Patterns. Mutations can be classified by their functional consequences. Mutations that inactivate the resulting protein, or prevent it from being made at all, are called loss-of-function mutations. These are usually recessive, since the organism still retains one functional copy on the other chromosome. Loss-of-function mutations may be dominant if the organism cannot compensate for the loss by using the other gene copy. Gain-of-function mutations are those in which the protein takes on a new function, or loses the ability to be regulated by other proteins. These mutations are typically dominant, since the new function may be deleterious even in the presence of a normal protein, encoded by the other gene copy.

Chromosomal Aberrations and Transposons

“Structural chromosomal aberrations,” the second category of mutations, arise when DNA in chromosomes is broken. The broken ends may remain unrepaired or may be joined with those of another break, to form new combinations of genes, such as translocations. A translocation between chromosomes 8 and 21 in humans causes acute myeloid leukemia by increasing the activity of *c-myc*, a gene involved in cell replication.

Translocations often cause human infertility, because they interfere with the normal distribution of chromosomes during meiosis. Chromosomes pair up before separating, as eggs or sperm are formed, and the correct pairing depends on matching sequences between them. Structural aberrations also include inversions and duplications of pieces of chromosomes.

Most chromosomal aberrations lead to the formation of chromosomal fragments without **centromeres**. Centromeres are crucial for proper chromosomal division, during both mitosis and meiosis. Therefore a chromosomal fragment is likely to be lost from one of the daughter cells formed after cell division.

Structural aberrations are nonetheless common in evolutionary history. As a result, although the chromosomes of mouse and man are quite different in appearance, most genes have the same neighbors in the two species, representing the ancestral mammalian arrangement, even if they have been moved to another chromosome as shown in Figure 3.

“Numerical chromosomal aberrations,” the third category of mutations, are changes in the number of chromosomes. In some cases, the whole genome has been duplicated (called **polyploidy**) and the mutant has, for example, four of each chromosome (and is thus tetraploid) rather than the usual two (diploid, as in humans). These are much more common in the evolution of plants than animals. In other cases, only one or a few of the chromosomes are involved, which is referred to as aneuploidy. Down syndrome, in which a person has an extra chromosome 21, is an example of such a mutation. Aneuploidy may also involve the loss of a chromosome. The absence of one of the sex chromosomes, X or Y, is a mutation in humans that results in Turner’s syndrome, in which there is only one X.

“Transposon-induced mutations” are the fourth category of mutations. Transposable genetic elements (transposons) are pieces of DNA that can copy themselves and insert into a new location in the genome. They were first discovered by Barbara McClintock, a U.S. geneticist and Nobel laureate in 1950. When transposons jump into a new position, the insertion may disrupt a gene and thus mutate it, usually inactivating it. Sometimes the transposon jumps again, and the activity of the gene it leaves is restored. Often, however, the transposon stays in the original position, permanently disrupting the gene. Some forms of hemophilia are due to transposon insertion. Transposon mutations have been extremely common in human evolution, and such mutations are still occurring.

Mutations in Research and Medicine

Early geneticists treasured mutations in the organisms they studied, since no characteristic can be studied genetically unless heritable variants exist. If, for example, everyone had brown eyes, nothing could be learned about the inheritance of eye color, as all generations would have the same color of eyes. For this reason, geneticists collected and propagated all the mutants they could find, and methods were developed to deliberately induce mutations, a process called mutagenesis. Such techniques include exposing their experimental organisms to X rays and chemicals.

Transposons can also be deliberately used to introduce mutations in model organisms. In the plant *Arabidopsis thaliana* and in the fruit fly

centromeres regions of the chromosome linking chromatids

polyploidy presence of multiple copies of the normal chromosome set



Drosophila melanogaster, transposon mutagenesis is often used to induce mutations, as the mutation can be very rapidly cloned and mapped with the transposon's DNA sequence as starting point.

Comparing existing mutations can help determine the evolutionary relatedness of two organisms. During evolution, there has been a relatively constant rate of accumulation of mutations in genes for a number of proteins, so the number of changes can be used to estimate the time since two species had a common ancestor. This is called the molecular clock and is illustrated in Figure 1. Each gene has evolved at a characteristic rate—the result of mutation rates, selection, and chance changes in the gene pool.

Advances in genetics have only intensified the search for mutations, especially in complex traits such as behavior and cancer, as the key to finding the genes involved and then unraveling the underlying mechanisms. This involves mapping the mutations, cloning the genes, and studying the mutants to discover what biochemical processes are changed in the mutants.

Mutations are believed to underlie most, if not all cancers. Cancer-causing mutations found so far include genes involved in communication between cells (signal transduction) and in the control of cell division. Many of these genes have been categorized into two broad classes: oncogenes and tumor suppressor genes. The mutation that has been found most often, in a tumor suppressor gene called *p53*, usually arises as a somatic mutation but can also be inherited as Li Fraumeni syndrome.

Xeroderma pigmentosum is an autosomal recessive condition in which the ability to repair DNA damage induced by UV light is defective. Many mutations are produced, and the affected people have large numbers of skin cancers. SEE ALSO CARCINOGENS; CHROMOSOMAL ABERRATIONS; DNA REPAIR; GENE; HEMOGLOBINOPATHIES; MUTAGENESIS; MUTATION RATE; POLYPLOIDY; TRANSPOSABLE GENETIC ELEMENTS.

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Bibliography

- Drake, John W. "Spontaneous Mutation." *Annual Review of Genetics* 25 (1991): 125–46.
- Hartwell, Leland H., et al. *Genetics: From Genes to Genomes*. Boston: McGraw-Hill, 2000.
- Lewis, Ricki. *Human Genetics: Concepts and Applications*, 4th ed. Boston: McGraw-Hill, 2001.
- Pauling, Linus, et al. "Sickle Cell Anemia, a Molecular Disease." *Science* 110 (1949): 543–548.

Internet Resource

International Agency for Research on Cancer. <<http://www.iarc.fr/>>.

Mutation Rate

Mutation rate refers to the frequency of new mutations per generation in an organism or a population. Mutation rates can be determined fairly precisely in experimental organisms with short generation times, such as bacteria or fruit flies. Human mutation rates are more difficult to determine accurately. Mutation rates can be used as a "molecular clock" to determine the time since two species diverged during their evolution.

Measurements of Mutation Rate

Mutation rate is often difficult to measure. The frequency of existing mutations in a population is not a good indication of the mutation rate, since a single mutation may be passed on to many offspring. In addition, there are often selective pressures that increase or decrease the frequency of a mutation in a population.

Mutation rates differ widely from one gene to another within an organism and between organisms. Generally mutation rates in bacteria are about one mutation per one hundred million genes per generation. While this sounds quite low, consider that the *Escherichia coli* bacteria in our intestines produce more than 20 billion new bacteria every day, each of which has approximately four thousand genes. This works out to about ten million new mutations in the population every day. In mice, the rate is about one mutation per ten thousand genes per generation. While this is much larger than the rate for bacteria, the mouse generation time is also much greater.

Human Mutation Rates

The appearance of rare dominant genetic diseases, such as retinoblastoma, have been used to estimate the mutation rate in the human population. Retinoblastoma is a childhood cancer of the eye and was a lethal condition until recently. Hence almost every case represented a new mutation (because individuals with the condition did not survive to reproduce and pass the genetic propensity for the disease along to their offspring), and the mutation rate could thus be readily estimated. Modern methods indicate that the mutation rate is roughly one mutation per 10,000 genes per generation. With at least 30,000 genes, this means that each person harbors about three new mutations, although this estimate may be off by a factor of ten. There are many more mutations in non-coding portions of DNA, but these are fairly difficult to study because they have no effect on the **phenotype** of the person.

About 90 percent of human mutations arise in the father rather than the mother. This may be related to the difference in the number of cell divisions required to produce a sperm versus an egg; sperm are produced late in a male's development, compared to eggs, which are produced quite early in the development of a female. Older parents pass on more mutations, and these may be either mutations within a gene or chromosomal aberrations, which are deletions or rearrangements of the chromosomes and involve many genes. Human mutation rates are generally quite similar worldwide. The exception is in local populations that have been exposed to radioactivity from nuclear testing or other sources.

Factors Influencing the Mutation Rate

Within a single organism, the mutation rate of two genes can differ by a thousandfold or more, so within a species some mutations may be very rare and others quite common. Exposures to very high doses of very potent mutagens can increase the mutation rate per generation by more than a hundredfold.

Both the nature of the gene and its environment can influence the mutation rate. The size of the gene, its base composition, its position in the

phenotype observable characteristics of an organism



promoter DNA sequence to which RNA polymerase binds to begin transcription

wild-type most common form of a trait in a population

base pair two nucleotides (either DNA or RNA) linked by weak bonds

enzymes proteins that control a reaction in a cell

genome, and whether or not it is being actively transcribed influence its mutation rate. The *dystrophin* gene, mutated in Duchenne muscular dystrophy, is thought to have a mutation rate of one in every ten thousand births, while the gene, mutated in Huntington's disease, has a mutation rate of closer to one in one million. The explanation for this difference is, at least in part, gene size: The dystrophin gene is one of the largest known. Genes whose **promoter** regions have been silenced by methylation (the addition of $-CH_3$ units to cytosine bases) are more likely to be mutated, since methylcytosine is easily converted to a base that resembles thymine.

In addition, the repair capacity of the organism can be important in determining how many mutations ultimately remain in the genome. For example, Bloom syndrome, a human cancer-causing condition, causes a decreased ability to repair DNA damage and an elevated mutation rate. Exposure to environmental mutagens or to protective agents, possibly dietary, can alter the mutation rate. Since the mutation rate is partly under genetic control, it is a selected characteristic of an organism, with the burden of detrimental mutations being balanced by the benefit of rare favorable mutations that are adaptive and permit evolution of the species.

One important factor influencing observed mutation rates is the means by which mutations are detected, as some methods may detect a changes at only one or two base pairs of a specific gene, leaving others undetected. Obviously the mutation rate observed by such methods will be lower than if more altered bases could be detected.

Mutations can also change a mutated gene back to the normal, **wild-type** form of the gene. Such "back" mutations are typically much rarer than "forward" mutations. This is because the number of ways to inactivate a gene is much greater than the number of ways to fix it. Imagine there are 1,000 bases that could be changed to produce a forward mutation. To reverse one of these mutations, it is necessary to change the one specific **base pair** that has mutated, and to change it back to the base it was before, rather than to a different one. Therefore, a back mutation rate of less than 1 one-thousandth of the forward rate would be expected.

The Origin of Spontaneous Mutations

The causes of most spontaneous mutations is not known, so the main factors affecting the spontaneous mutation rates are obscure. It is likely that the methylation of cytosine in the DNA is important for many spontaneous mutations, because many are found at sites in the genes where cytosine is methylated. Ionizing radiations, such as cosmic rays, probably account for less than 10 percent of spontaneous mutations. Other factors are errors made during replication of the DNA; exposure to mutagens produced by cells during their normal metabolic activity, with reactive oxygen species being the common suspect; spontaneous breakdown of DNA at body temperatures; and exposure to environmental agents. Many mutations are made when the mechanisms that repair DNA make mistakes, and many error-prone DNA repair **enzymes** are known.

Molecular Clocks

While it may seem unlikely, it is believed that the overall mutation rate within a species does not vary much over long periods of time. This means

that the mutation rate serves as a “molecular clock.” The clock can be used to determine the time since the evolutionary divergence of two species. Two organisms with very few DNA sequence differences between them diverged more recently than two that display more accumulated differences. The absolute amount of time for these divergences can be determined if the clock is calibrated, that is, if a known number of sequence differences can be correlated with a known time since divergence. This is done by comparing sequence data with data from the fossil record.

Early work in this field concentrated on a small handful of genes and gave conflicting results. Because mutation rates vary among genes, the best results come from analyzing changes in many genes. A 1998 study of the evolution of mammals analyzed 658 nuclear genes from 207 vertebrate species. It showed that the ancestors of most contemporary mammals arose more than 80 to 110 million years ago, long before the extinction of dinosaurs, and demonstrated that the fossil record from that time has some very large gaps in it. SEE ALSO CARCINOGENS; DNA REPAIR; GENE; METHYLATION; MOLECULAR ANTHROPOLOGY; MUSCULAR DYSTROPHY; MUTAGEN; MUTATION.

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Bibliography

Crow, J. F. “The High Spontaneous Mutation Rate: Is It a Health Risk?” *Proceedings of the National Academy of Science* 94 (1997): 8380–8386.

Griffiths, Anthony J. F., et al. *An Introduction to Genetic Analysis*. New York: W. H. Freeman, 2000.

Kumar, Sudhir, and S. Blair Hedges. “A Molecular Timescale for Vertebrate Evolution.” *Nature* 392 (1998): 917–920.

Nature of the Gene, History

Although Wilhelm Johannsen coined the term “gene” in 1909, our understanding of the nature of the gene has changed significantly over the course of the twentieth century. Gregor Mendel’s elements of inheritance were given a material basis in the chromosome theory of the early twentieth century. Attempts to understand the nature of gene action and mutation spurred interest in the biochemical role and molecular basis of the gene, culminating in the discovery of the structure of DNA.

From Elements to Genes

In 1865, when Mendel articulated the laws of inheritance that now bear his name, he did not use the terms “gene” or “**allele**.” He referred instead to “elements” and “characters.” Mendel described the patterns of inheritance he observed in terms of character pairs. These pairs segregate to form the next generation of character pairs and remain independent of the behavior of other character pairs. The external characters of the pea plants he described corresponded to elements within the **germ cells** of the same plants. Whether Mendel thought that pairs of characters were expressions of pairs of cellular elements is not clear.

allele a particular form of a gene

germ cells cells creating eggs or sperm

